Basics of Spark

Wednesday, 26 August 2026 6:57 PM

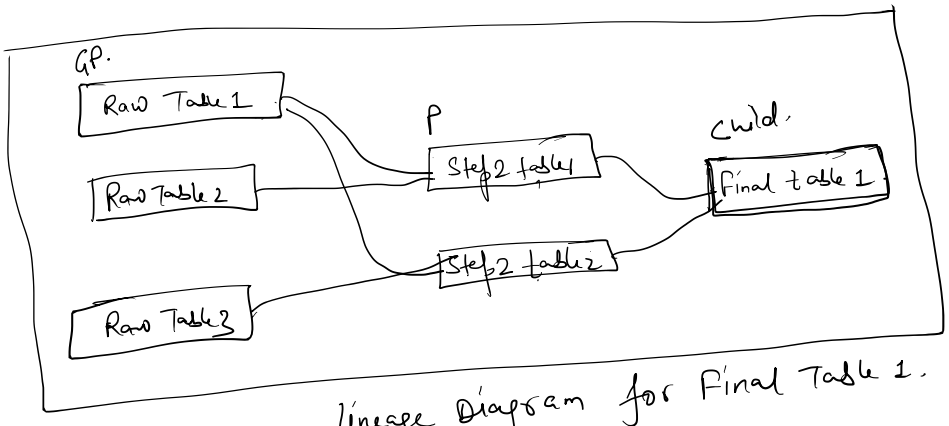
Hadoop version 1.x → default processing engine was MR.

Spark: It is an open source, distributed processing engine designed for fast and flexible analytics on large datasets.

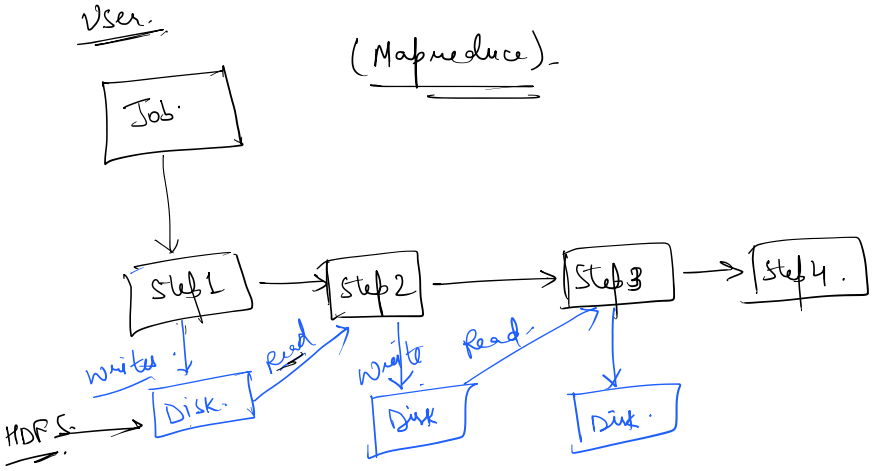
Supported in Scala, Java and Python.

Hadoop version 2.x onwards → spark is supported.

Feature	MR	Spark.
Data Processing	Disk Based → Read/Write to disk	In Memory → keeps intermediate results in RAM.
Processing Speed	Slower → Due to multiple disk read/write	Faster → due to in-memory processing.
Programming Difficulty	Requires long/verbose Java Code.	provides simple/high level API in scala/Java/Python.
Fault Tolerance	Lesser writes all intermediate results to disk.	uses RDD lineage to remember the instructions.
Execution Flow	Step by step → Sequential	Flexible → Directed Acyclic Graph (DAG).
Streaming/Real time.	Does not support Real time or streaming	Supports near real time.
Hardware Requirements	Lesser RAM needed	High RAM requirement.



Lineage Diagram for Final Table 1.



* RDD (Resilient Distributed Datasets)

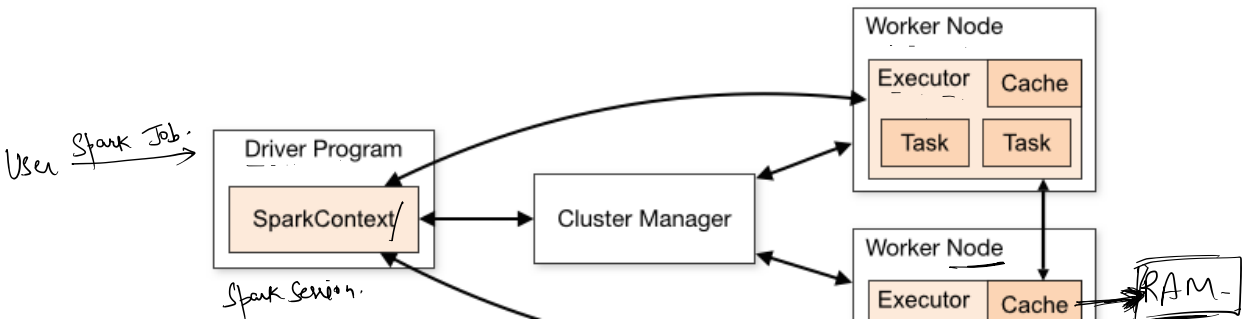
1) Resilient: Fault tolerant

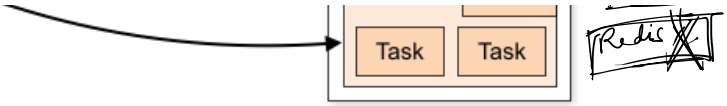
- If a machine crashes / data is lost, spark does not have to go back and restart from scratch.
- It simply looks at the lineage and just recomputes the missing chunk.

2) Distributed: - Data is not on one computer.
- It is partitioned and stored cross RAM of different systems.

3) Dataset: Collection of Records.

* Components of Spark:





1) Driver Program : This take the python / scala code and creates the actual execution plan.

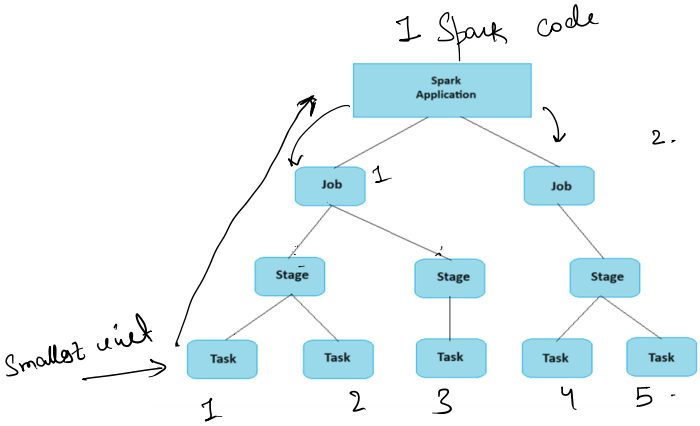
2) SparkSession (earlier SparkContext) : It allows the code to interact with the Spark cluster. Also, does the configuration setting for the job.

3) Cluster Manager : It is responsible for managing the resources
Spark Standalone / YARN / → Yet Another Resource Negotiator.
Kubernets. It is the one which reserves CPU / Memory for the Job-

4) Worker Node : A physical server / VM, in your cluster.

5) Executor : A spark process running inside the worker node

6) Task : Smallest unit of work in Spark,

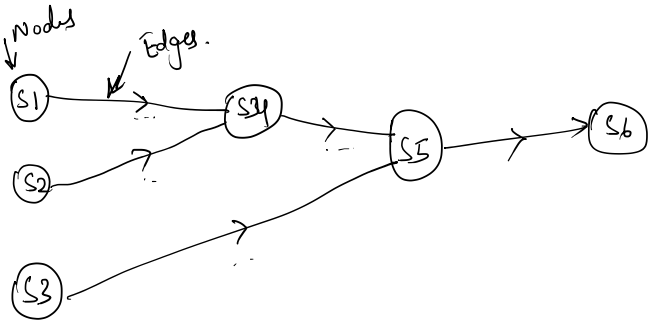


* DAG (Directed Acyclic Graph).

1) Directed → It means it flows in a direction, i.e. forward,

→ No loops, No cycles.

2) Acyclic



Use Cases:

SparkML → ML applications using Spark.

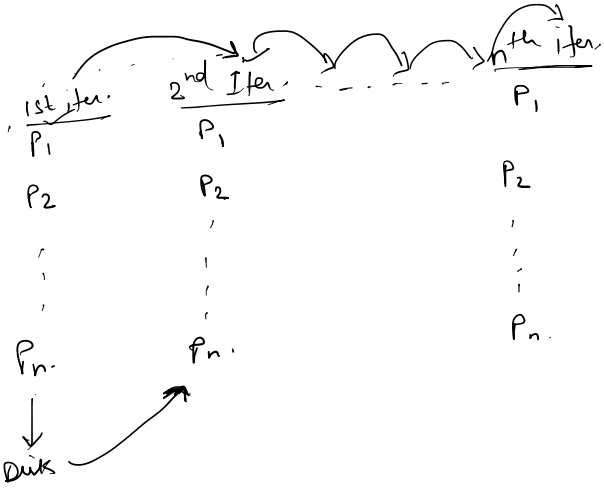
1) Machine learning:

Feature calculation is done on historical data.

House Price Prediction.
Past 5 years data of houses → Attributes & price

C1	C2	C3	C4	Y

Data → ML model learning →



2) Recommender Systems. (Spark)

Recommended for You.

↳ Uses a lot of historical data.

Training a recommender system.

Spark → 0.5 to 1 sec

* Lazy Evaluation:

- In spark, it does not execute the command as soon as you

submit it.

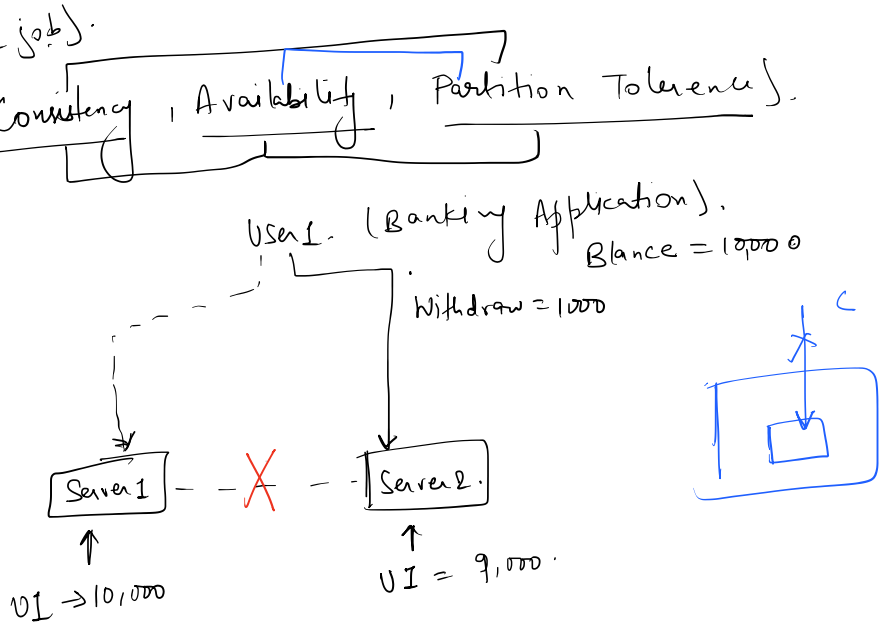
— It will execute the job if you give any action command.

Print, write to disk.

sales_job = "Give me sales of last 5 years → avg sales per month"

print(sales_job).

* CAP (Consistency, Availability, Partition Tolerence).



Scenario 1 → Sorry, try again later.

Consistency + Partition Tolerence. (CA)

S2 S1

Scenario2 → Return the balance as 9K/10K.

Availability + Partition Tolerence. (CP)

8GB → 1TB.

RAM

8GB +

50MB → 2GB

128GB

128GB.

Hard disk.

1000GB.